

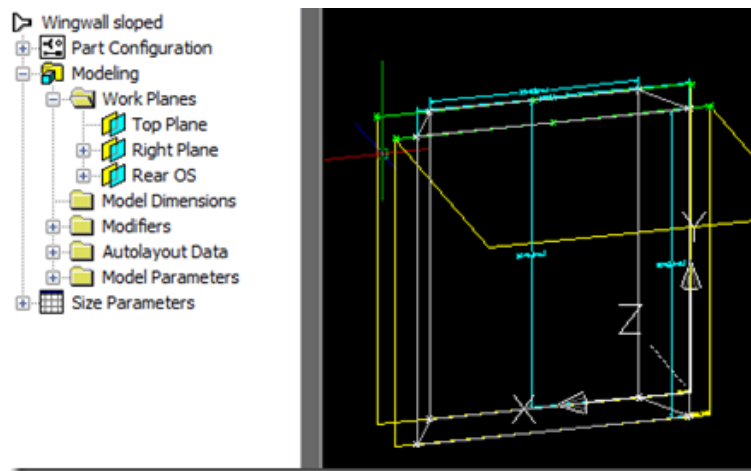
CIVIL 3D – PART BUILDER PART 4

Posted by John Evans | Dec 9, 2008 | AutoCAD Civil 3D

Source : <https://designandmotion.net/autodesk/autocad-civil-3d/civil-3d-part-builder-part-4/>

Introduction

We will pick up where we left off from Part Builder Part 3. Open our saved part “Wingwall sloped” from the last session. Adjust the view so that you have something similar to the following image:



It's time to speed things up a bit, so as I refer to procedures, I will once again detail them, and thereafter I will just state that it needs to be completed.

Note: Halfway through I realized some changes needed to be made, in order for this process not to be more confusing than it might already appear. If I missed any odd things in the images, or got out of sync with the variable names, please notify me, and I'll do my best to correct them.

In this session we will create another offset plane, and additional profiles and transitions. Before we do, it would be prudent to create the remaining profiles that will be referenced to our (current) Right Plane. These will be the locations that the extending walls project into, the “tie in” points. You should be fairly familiar with the basic procedures, *Lord knows you've had tons of time to practice since my last post*, we will pick up the speed a bit.

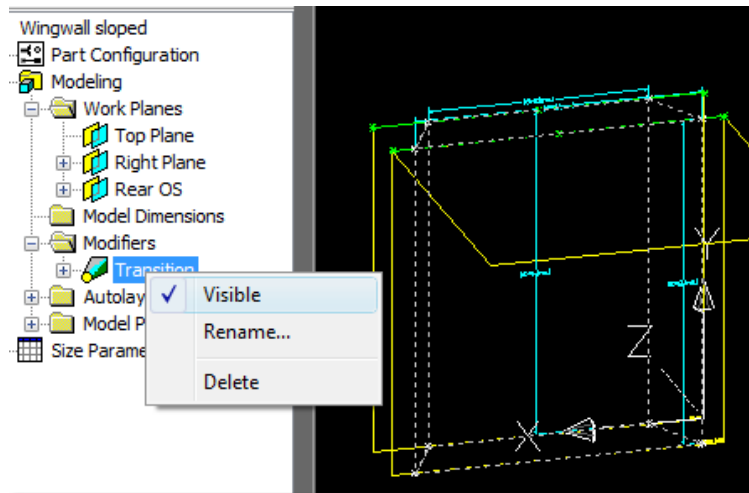
Note: I will refer to the profiles by Name, to avoid confusion. Some have not been created yet, so you will understand better as time goes on. I felt that telling you NOW instead of in the middle of this was better for your understanding. I will show them in UPPERCASE as well. The names are as follows:

- BASE PROFILE = the profile controlling the bottom Transitions, the “floor”.
- LEFT and RIGHT PROFILE = These are the wing wall profiles, symmetrically placed to the Left and Right on the respective planes.
- ORIGINAL PROFILE = This is the Profiles already created in the previous sessions, that represent the Rear Wall.

VISIBILITY

Sometimes you have numerous features lying atop one another, and some visibility is really needed. Selection control is reduced, so shutting things on and off is essential, as you will see. Let's shut off the extrusion so that we can constrain to the previous Profile. Otherwise when we try and select things, we get the last thing created.

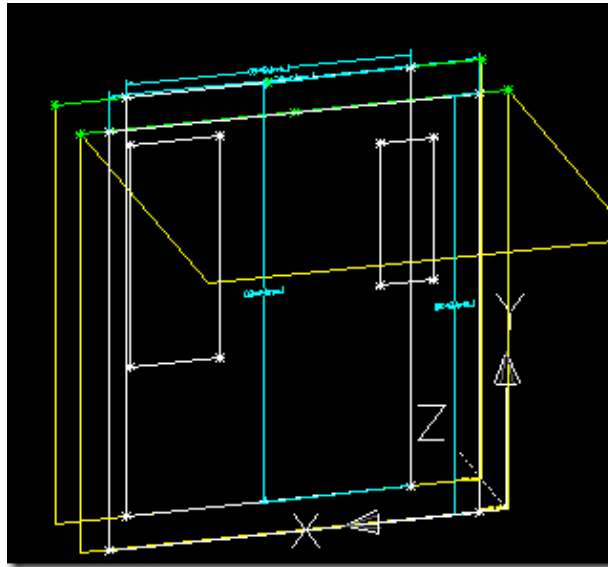
Expand the Modifiers, and Right click on the Transition, and Select Visible.



As you should see, the Extrusion disappears, leaving the 2 ORIGINAL PROFILES previously created. Now we can add 2 rectangular Profiles to the Right Plane, one to the RIGHT, and one to the LEFT, lying inside the ORIGINAL PROFILE that was already there.

RIGHT PLANE REVISITED

Right click on the Right plane, Select 'Add Profile', and then 'Rectangular...'. Select two points inside the ORIGINAL PROFILE, and repeat the process. Here is a sample to give you some reference.



Now we need to dimension and constrain the new Profiles. Add dimensions to the LEFT and RIGHT PROFILE for the width. We will not need the height dimensions, as we will control the height with constraints. Right click on the Right plane, Select 'Add Dimension', then 'Distance...'. Select the two Top points of the LEFT PROFILE, and repeat the process for the RIGHT PROFILE as well.

[*TIP: The menu scripts commands to the command line, so you can right click ENTER, or just hit the ENTER key to reinitiate the command in repetitive processes.]

We want these LEFT and RIGHT PROFILES constrained to the ORIGINAL PROFILE. Constrain the upper left point of the LEFT profile, coincident to the upper left point of the ORIGINAL PROFILE, then the upper right point of the RIGHT PROFILE coincident to the upper right point of the ORIGINAL PROFILE.

Right click on the Right plane, Select 'Add Constraint', and then 'Coincident...'. Select the point combination as specified above. The bottom of our walls are still hanging free, and need to be tied to a base.

The wall we have created so far has no base. I had considered adding the base at the end of the workflow, but I fear that the additional constraints will be too confusing. To simplify the model, and this process, we will assemble the base into the face of the Right Plane. To do that, we need to add another rectangular profile below the geometry we have now, with a height of 'bThick' Hereafter, this will be called the "BASE PROFILE".

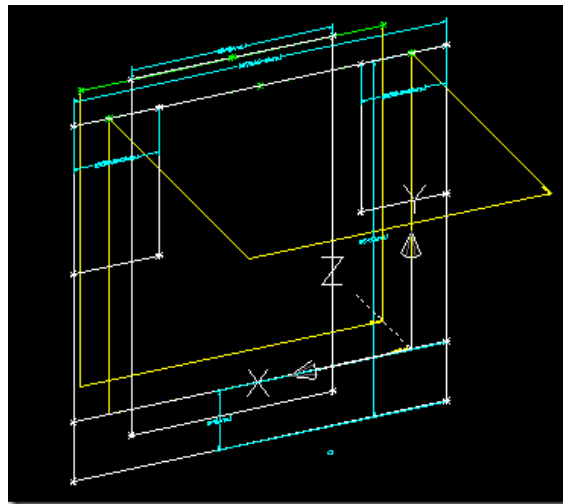
Add a Rectangular Profile to the Right plane, selecting two points inside the previous Profile, near the bottom. Add a distance dimension to the points along the right side of the BASE PROFILE. Now we need to constrain it to the bottom of the face of the wall (ORIGINAL PROFILE). Add a Constraint to the lower points of the BASE PROFILE, Coincident to the lower points of the ORIGINAL PROFILE, Left to left, and right to right.

Now we need to add the thickness to the BASE PROFILE, Add the angle parameter, and edit the dimensions of the 2 LEFT and RIGHT PROFILES. Right click on the Model Parameters collection, and select 'Edit...'. Double click on the LenA7 value under the equation column, and change the value to be 'bThick'.

Now it is very important to ADD THE THICKNESS to the bottom of the REAR WALL TO ACCOMODATE THE NEW BASE.

- LenA2 (Right Plane height dim) = 'bHeight + bThick'
 - LenA4 (the Rear OS plane height dim) = 'bHeight + bThick'
 - Select the Add button and create a new User Parameter, called 'bAng', set the equation to be 45, and the custom parameter to degrees.
 - LenA5 = 'bThick/cos(bAng)'
 - LenA6 = 'bThick/cos(bAng)'
- Select the Close button.

You should see something that looks similar to this. Some dimensions were turned off for clarity.



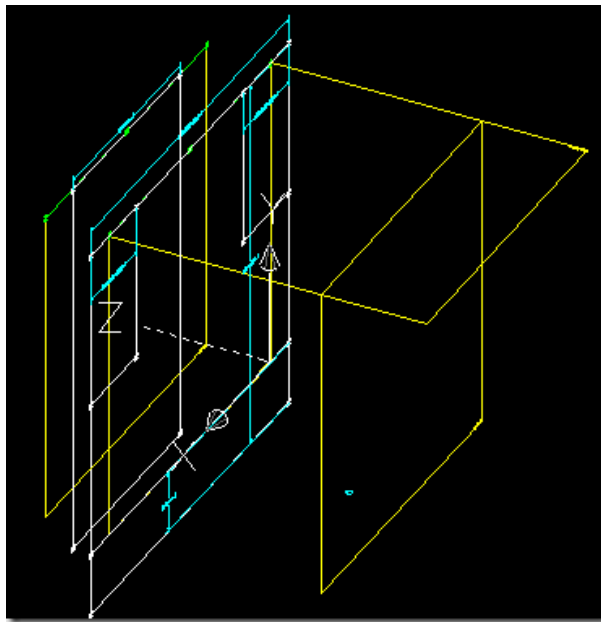
Now that the base is positioned, ALL WALLS will reference the BASE PROFILE. This consistency will make the model easier to understand and easier to troubleshoot. Lets constrain the LEFT and RIGHT PROFILES down to the BASE PROFILE.

Add a Constraint referencing the lower left point of the LEFT PROFILE, Coincident to the upper left point of the BASE PROFILE. Repeat this process, on the right side of the RIGHT PROFILE. The walls should be constrained and all the dimensions set. Now we need an offset plane, to control the middle of the part.

THE MID PLANE

Do not become frustrated by the equations I use in this session. They are not complex, but a bit long occasionally. Don't worry. This is not because of Part Builder, and you may never use this kind of feature in your work. I needed to affect offsets based on the angles we are using, so that no matter what slopes or widths, the part will always shape out properly. For now, just type out the equations. If you use the calculator to key them in, then make sure to evaluate the extra parenthesis before you close the dialog.

Right Click the Work Planes folder and select 'Add Work Plane...'. Select 'Offset', Enter a description (I used "Mid Plane") and click OK.



Pull the plane towards you in the view and select a point to set the offset distance. It doesn't matter what distance, as long as you can still see the plane. We will change the distance in the parameters. Lets go there and correct the OS plane distance, as well as the descriptions of the OS Plane Distances. Right click on the Model Parameters collection, and select 'Edit...'

- WPOf2 = 'bWidth'...Change the description to "Mid Plane OS Distance".
- WPOf1 = 'bThick'...Change the description to "Rear OS Distance".

A value from Part 3 needs to be updated. The LenA1 parameter/dimension needs to reflect the Width of the structure, PLUS the angular cross section of two walls.

$$\text{LenA1} = \text{'bWidth} + ((\text{WPOf1} / \cos(\text{bAng})) * 2) \text{'}$$

Close the dialog.

[*Tip: Use the calculator button for the complex (sort of) equations, and press the evaluate button to see if what you enter is what you expect before proceeding.]

Now we have to repeat the anchoring process outlined in the two previous planes. I will simply say to go back to Part 3 and review the anchoring process for the Rear OS plane, and repeat it EXACTLY for the Mid Plane. Anchor the plane features as previously discussed for the Rear OS plane. Add the BASE PROFILE at the bottom for the base, and add two distance dimensions, one along the top points, and one along the side points. Add a distance Dimension from the Green anchor line, to the bottom line of the new BASE PROFILE.

We tied the dimension to the bottom because we want all the dimensions to be similar for each plane to plane, making troubleshooting easier. Constrain the new BASE PROFILE such that the outer bottom points are equal distances from the mid anchor point. At this point the New BASE PROFILE should have two dimensions controlling its size, and constraints and a dimension tying its location. Add LEFT and RIGHT rectangular profiles above the BASE PROFILE, and add height and width dimensions to the LEFT PROFILE only.

We will use constraints this time to control the RIGHT PROFILE without having to dimension it, such that as the LEFT PROFILE parameters are changed, the RIGHT PROFILE will update in kind (we could just dimension the RIGHT PROFILE as well, but I want you to practice the Equal Distance Constraints). Add Equal Distance Constraint to the RIGHT PROFILE, so that the distance along its upper points is equal to the distance along the upper points of the LEFT PROFILE. Repeat the constraint for the side distance.

When the model updates, the profiles should be identically sized. Now let's get them fixed in space. Constrain the LEFT PROFILE by its lower left point, coincident to the upper left point of the BASE PROFILE. Repeat the process for the lower right point of the RIGHT PROFILE, to the upper right point of the BASE PROFILE.

Now that the features are controlled in space, let's change the dimensions. Edit the Model Parameters as follows:

- LenA8 (controlled height) = 'bHeight + bThick'
- LenA9 (Base width) = ' $\text{LenA1} + ((\tan(b\text{Ang}) * WPOf2) * 2)$ '
- LenA10 (Base thickness) = 'bThick'
- LenA11 (left width) = 'bThick'
- LenA12 (left height) = no changes yet.

Below is a screenshot of the variables as I have them now, just in case something got fouled up. Remember, it doesn't matter what your variable names are, as long as the equations are related to the correct geometry.

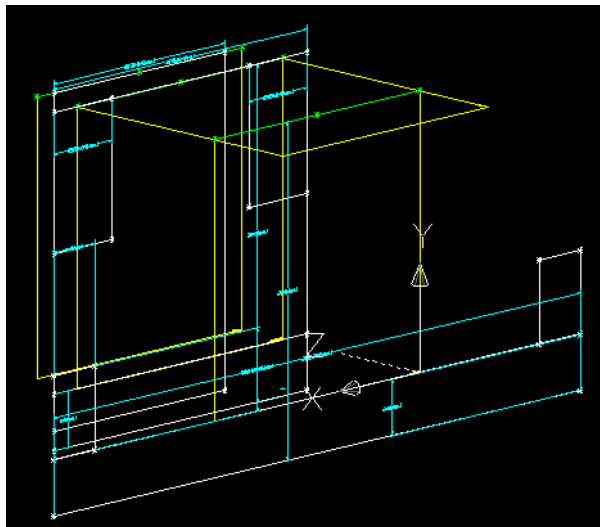
Name	Value	Equation	Description
bAng	45.0000	45	Model Angle Parameter 1
bHeight	30.0000	30	The Wall's Base Height (at Rear)
bThick	6.0000	6	Model Distance Parameter 3
bWidth	20.0000	20	The Wall's Base Width (at rear)
LenA1	36.9706	$bWidth + ((WPOf1 / \cos(bAng)) * 2)$	Right Width
LenA10	6.0000	bThick	2D Distance 10
LenA11	6.0000	bThick	2D Distance 11
LenA12	8.9413	8.94125	2D Distance 12
LenA2	36.0000	bHeight+bThick	Right Height
LenA3	24.9706	$bWidth + ((bThick / \cos(bAng)) - (WPOf1 * \tan(bAng))) * 2$	Rear OS Wdth
LenA4	36.0000	bHeight+bThick	Rear OS Height
LenA5	8.4853	$bThick / \cos(bAng)$	2D Distance 5
LenA6	8.4853	$bThick / \cos(bAng)$	2D Distance 6
LenA7	6.0000	bThick	2D Distance 7
LenA8	36.0000	bHeight+bThick	2D Distance 8
LenA9	76.9706	$LenA1 + ((\tan(bAng) * WPOf2) * 2)$	2D Distance 9
SHBTh	0.0000	0.000000	Headwall Base Thickness
SHBW	0.0000	0.000000	Headwall Base Width
SVPC	0.0000	0.000000	Vertical Pipe Clearance
WPOf1	6.0000	6	Rear OS Distance
WPOf2	20.0000	bWidth	Mid Plane OS Distance

Oh, I forgot to change the rear plane offset (you may have already changed this value):

- WPOf1 (Rear OS Distance) = 'bThick'

Close the dialog.

Here is the litmus test. If you rotate your view such that the left edge of all LEFT PROFILES, ORIGINAL PROFILES, and BASE PROFILES are aligned. There should be nothing out of line.

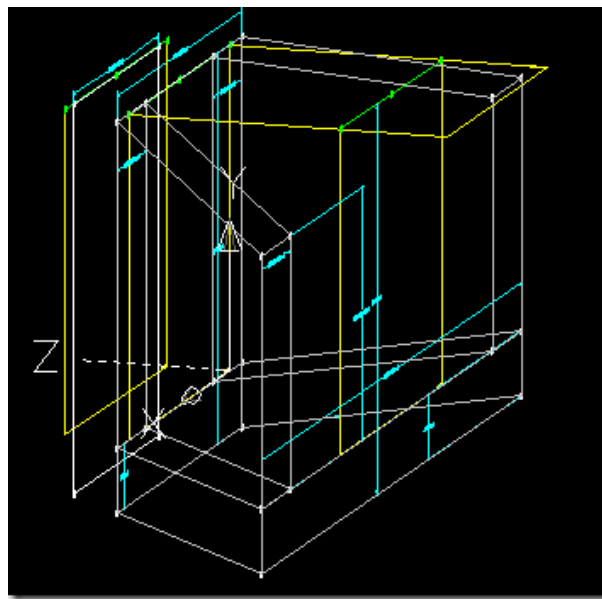


This is the milestone. After this, it's easy. We need to set the height of the walls at this plane. This wall is sloped according to grade, and we want that to be controllable. We need a User Variable. Create a User Variable named 'bSlope', with a value of 3.

- Change LenA12 = 'bHeight-(WPOf2/bSlope)'

Now we can create Transitions between the profiles on the Mid plane and the respective profiles on the Right Plane. Right Click 'Modifiers', and select 'Add Transition...'. Select the BASE PROFILE on the Mid Plane, and then the BASE PROFILE on the Right Plane. Repeat for the LEFT PROFILES, and then the RIGHT PROFILES.

Don't forget you can ENTER to repeat the last command. You should end up with something like this.



Did you ever get homework from a workflow session? Let me be the first.

Create the Toe Plane, Offset from the Right Plane a distance of $(bHeight - bThick) * bSlope$. Insert the Anchors. Create the appropriate BASE, LEFT, and RIGHT PROFILES, dimension and constrain them properly. The Width and Height of the LEFT PROFILE is 'bThick' (You decide how to get the RIGHT PROFILE sized properly). The Width of the BASE PROFILE is the width of the Mid Plane BASE PROFILE. Add the 3 Transitions. You'll get no credit if you can't create the Transitions because you could not select the Mid Plane Profiles. (That's 2 hints in 1, by the way)

Remember when using parameters that can feed off of each other, to use the variable from the reference, and not to copy its equation. For example, previously we needed the equation from LenA1 to complete the calculations for LenA9...LenA9 = $'LenA1 + ((\tan(bAng) * WPOf2) * 2)'$... We simply used LenA1 to carry all that over. This way everything cascades.

[Tip*: any time you can change 1 parameter and effect numerous others intentionally, then DO IT]

In the next session we will review your Homework. We will go on to create the preview image, and set the layout point. We will configure all the Table data, and publish this thing.